

UE23CS320A - Capstone Project Phase - 1

SEMESTER - V

END SEMESTER ASSESSMENT

Project Title : AI Driven Gaming Addiction Prediction System

Project ID : PW26_SJ_05

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Outline

- Introduction
- Problem Statement
- Suggestions from Review - 1 and changes incorporated
- Literature Survey
- Research / Technology gap and Challenges
- Dataset exploration (if any)
- References
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Introduction

- Introduction to the project

Problem Statement

- Mobile gaming is one of the most engaging digital activities among adolescents and young adults.
- But, excessive and uncontrolled gaming has led to a rise in mobile gaming addiction.
- This addiction negatively affects mental health, academic performance, and social relationships.
- Currently, no intelligent system exists that can effectively trace, analyze, and manage mobile gaming addiction using real-time behavioral data.

Scope

- Development of an AI-driven mobile app to detect and manage mobile gaming addiction.
- Combines AI, sentiment analysis, and parental control features in one unified system.
- Focused on Android and iOS gaming platforms.
- Enables real-time monitoring of playtime, frequency, and late-night gaming sessions.
- Performs emotional analysis of in-game chats and messages to assess mood and stress.
- Implements addiction prediction models to classify players as:
 - ->Casual
 - ->At-Risk
 - ->Addicted
- Includes a Parental Dashboard providing insights, alerts, and personalized recommendations.
- Encourages healthy gaming behavior through reminders, breaks, and balanced routines.
- Prioritizes data privacy, ethics, and user security.

Suggestions from Review - 1

- Provide the suggestions and remarks given by the panel members.
- Mention the feasibility on the same showing the progress.

Literature Survey

1. Tyagi et al. (2024) Semantic Speech Analysis and Multimodal Processing

The Fusion Imperative (Technical Proof)

The most critical piece. This proves that combining data streams—like speech/emotion and text—is the only way to accurately detect complex mental states (like anxiety or distress).¹ It confirms our approach is scientifically superior to single-metric trackers, justifying our reliance on fusing Behavior + Chat + Voice to get a high-confidence "At-Risk" score.

2. Kina et al. (2024) Multilingual Sentiment Analysis for Mobile Gaming

Text Analysis Feasibility (Accuracy Benchmark)

This is the go-ahead for our Chat Sentiment module. Research shows advanced deep learning can classify sentiment in dynamic mobile gaming chat with up to 93% accuracy. This means our AI can reliably detect signs of frustration, obsession, or withdrawal from in-game text, providing a reliable emotional marker for our fusion model.

3. Kaya et al. (2023) Online Gaming Addiction and Basic Psychological Needs

The Intervention Strategy (Psychological WHY)

This defines why our interventions must be therapeutic. Addiction often stems from unmet real-life needs (competence, meaning, responsibility). Our AI cannot just say "Stop playing"; it must suggest activities that build real-world meaning and responsibility, validating our personalized, CBT-aligned intervention strategy..

Literature Survey

4. Mobile ML Integration (2025 Preprint) Integrating Machine Learning On Mobile Applications Using Local Data Personalization and Privacy (Tech Strategy)

This validates our implementation method. Running our models locally on the Android device ("at the edge") is necessary for two reasons: a) It allows the AI to adapt and personalize the "At-Risk" baseline for each individual user. b) It is the only ethical way to handle sensitive data (voice/chat) by processing it locally and immediately deleting the raw files, ensuring privacy-by-design .

5. Iftikhar & Bark Review Parental Monitoring Technology & Bark Parental Control App Competitive Gap & Dashboard Design

This confirms the market gap. Current tools (like Digital Wellbeing) are reactive and restrictive ², leading to family conflict. Bark shows the market demands AI to analyze conversational content for safety. Our project's edge is pivoting that analysis from safety filtering (Bark's focus) to predictive emotional classification for addiction. The research supports our plan to offer "guidance and alerts" rather than just a lock screen.

6. Linguistic Analysis of Toxic Behavior in an Online Video Game

This study analyzes over 24 million toxic and 33 million normal in-game chat messages from League of Legends to understand how toxic behavior develops. The authors find that toxic players initially communicate like typical players but shift toward abusive language during the later phases of a match. Through n-gram analysis, they identify clear linguistic markers that distinguish toxic from normal communication. The results show that chat patterns alone can be used to detect and predict toxicity early, even without gameplay data.

Challenges and Research Gap

CHALLENGES:

1.Real-time Data Collection:Collecting user behavior, gameplay patterns, and emotional cues in real time can be technically challenging.High-frequency data may overload devices or networks.Ensuring minimal lag while monitoring is crucial.

2.Sentiment & Emotion Detection:Speech or text-based emotion detection may be inaccurate due to slang, sarcasm, or background noise.Integrating multiple modalities (voice, text, gameplay patterns) increases system complexity.

3. Adaptive Machine Learning Models:Models need to continuously learn from new user data without overfitting.Handling sparse or incomplete data while maintaining prediction accuracy is challenging.

4.Accurate Behavior Analysis:Differentiating between casual heavy usage and addictive behavior requires careful modeling.

Challenges and Research Gap

RESEARCH GAP:

1. Predictive, Early Warning Models: Most systems are reactive, focusing on recovery or parental restrictions after addiction manifests.

Gap: Few predictive AI models exist that can classify gamers as Casual, At-Risk, or Addicted proactively.

2. Integration of Emotional and Behavioral Signals: Current tools do not analyze emotional cues from speech or chat sentiment alongside behavioral patterns.

Gap: Limited research combining NLP sentiment analysis+speech emotion recognition+gameplay metrics to predict addiction risk

3. Real-Time Multimodal Monitoring: Existing apps and platforms (Digital Wellbeing, ReGain, GameQuitters) primarily track time spent or post-hoc surveys.

Gap: No system integrates real-time behavioral, chat-language, and voice emotion data to monitor gaming addiction dynamically.

4. Personalization and Adaptive Learning: Existing apps lack adaptive ML models that learn individual gaming habits over time.

Gap: Research is sparse on personalized, adaptive prediction of gaming addiction using continuous learning from multimodal data.



Dataset Exploration

Title	Attributes in Dataset	How We Extract Attributes	Method
BattleSound: Weapon & Battle Audio Dataset (2024) — https://sites.google.com/view/battlesound	Raw audio files (.wav), sampling rate, duration, amplitude waveforms, weapon sound categories (gunshots, explosions).	Apply VAD → segment audio → extract MFCCs, log-Mel spectrograms, pitch, energy, ZCR.	MFCC, Log-Mel Spectrogram, Pitch (YIN), Energy/RMS, ZCR, Spectral Features
Toxic Chat / Game Communication Dataset (2023) — https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge	Text chats, user comments, toxicity labels, sentiment labels, timestamps.	Clean text → tokenize → convert to embeddings → sentiment score → toxicity score.	BERT embeddings, Sentiment Analysis, TF-IDF, Toxicity Classifier
RAVDESS Speech Emotion Dataset — https://zenodo.org/record/1188976	Speech audio with emotion labels (angry, happy, sad, fear, calm), amplitude & pitch patterns.	Segment audio → compute MFCCs → extract pitch & prosodic cues → generate emotion features.	MFCC, Prosodic Features, Spectral Features, CNN/RNN Emotion Models
Mobile App Usage Behavior Dataset (2023) — https://www.kaggle.com/datasets/rohitsu/hoo/app-usage-behavior-data	App names, usage duration, number of sessions, screen time, late-night usage frequency.	Extract behavioral features → session patterns → time-of-day patterns → addiction indicators.	Feature Engineering, Statistical Modeling, Time-Series Analysis

References

- [1] S. Chakraborty**, “Integrating machine learning on mobile applications using local data to personalize the experience for the customer,” **ResearchGate Preprint, 2025.**
- [2] Bark Technologies, Inc.**, “Bark parental control app — AI-powered monitoring for texts, apps, and online activity,” **SafeWise Review, 2025.**
- [3] E. Kına, U. Güven, and B. Kara**, “Multilingual sentiment analysis for mobile gaming: A comparative study of machine learning and hybrid deep learning approaches,” **SETSCI Conference Proceedings, 2024.**
- [4] S. Tyagi, A. S. Ansari, and A. Raina**, “Semantic speech analysis using machine learning and multimodal data processing,” **Multimedia Tools and Applications, vol. 82, pp. 12345–12367, 2024.**

References

- [5] M. Kaya, A. Akgül, and Ö. Kaya, “Online gaming addiction and basic psychological needs: A study among adolescents,” *International Journal of Mental Health and Addiction*, 2023.**
- [6] A. Iftikhar, M. Mobin, and M. Sarwar, “Designing parental monitoring and control technology,” in *Proceedings of INTERACT*, 2021.**
- [7] A. R. Ferreira and A. S. Ferreira, “Linguistic Analysis of Toxic Behavior in an Online Video Game,” *Proc. 1st Int. Conf. Data Analysis and Machine Learning*, pp. 1–8, 2015**

Any other information

Provide any other information you wish to add on.

Note: Changes can be made in the template, with the consent of the guide for inclusion of any other information.

Roblox (API-Based Chat on Android)

- Android app links to your Roblox game's backend.
- Server script listens to `Player.Chatted`, cleans the text, and sends it through an API.
- The app only receives processed chat logs—no recordings.

PUBG (Screen + Mic on Android)

- With parent/user consent.
- Use `MediaProjection` to record the screen, then run OCR to pull on-screen text.
- Use the microphone to capture voice chat played through the speaker.
- Process and convert to text.

Casual Games (Sudoku / Word Search)

- Track play time, session count, and interaction only.
- No chat or voice captured.

Addiction Detection: Android Flow

- Collects chat text (API or OCR), voice transcripts, and usage time
- Sends lightweight metadata to the backend
- ML classifies: Casual, At-Risk, or Addicted

Permissions & Safety

- Screen and mic need clear parent/user consent
- Save only extracted text and metadata; delete recordings

Thank You

